Tables

1 Consider the function $f(x) = \dfrac{1}{7-x}$.

a Complete the first table of values, in which $x < 7$

b Complete the second table of values, in which $x > 7$

c Find the limit of $f(x)$ as the value of x approaches 7.

x	5	6	6.9	6.99
$f(x)$				

x	9	8	7.1	7.01
$f(x)$				

2 Consider the function $f(x) = 5x^2 + 1$:

a Complete the table to find the exact values of $f(x)$ as x gets closer and closer to 2 from the left, and closer and closer to 2 from the right:

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$						

b Find the value of $\lim_{x \to 2} \left(5x^2 + 1\right)$.

3 Consider the function $f(x) = \dfrac{2-x}{x^2+2}$:

a Complete the table to find the values of $f(x)$ as x gets closer and closer to 0 from the left, and closer and closer to 0 from the right. Round your answers to four decimal places.

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$						

b Find the value of $\lim_{x \to 0} \left(\dfrac{2-x}{x^2+2}\right)$.

4 Consider the function $f(x) = \dfrac{x^2 - 4x}{x-4}$.

a Complete the table to find the values of $f(x)$ as x gets closer and closer to 4 from the left, and closer and closer to 4 from the right:

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$						

b Find the value of $\lim_{x \to 4} \left(\dfrac{x^2 - 4x}{\quad}\right)$.

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5 Consider the function $f(x) = \dfrac{x^3 + x + 2}{x + 1}$.

a Complete the following table:

x	-1.1	-1.01	-1.001	-0.999	-0.99	-0.9
$f(x)$						

b Find $\lim_{x \to -1} f(x)$.

6 Consider the function $f(x) = \dfrac{\sqrt{x} + 4}{x - 5}$:

a Complete the following table, rounding all values to two decimal places:

x	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$						

b Does $\lim_{x \to 5} f(x)$ exist? Explain your answer.

7 Consider the limit: $\lim_{x \to -5} \left(\dfrac{x^2 + 4}{x + 5} \right)$.

a Complete the table.

x	-5.1	-5.01	-5.001	-5	-4.999	-4.99	-4.9
$\dfrac{x^2 + 4}{x + 5}$				–			

b Does the above limit exist? Explain your answer.

8 Consider the limit: $\lim_{x \to 3} \left(\dfrac{e^{x-3} + x - 4}{x - 3} \right)$.

a Complete the following table:

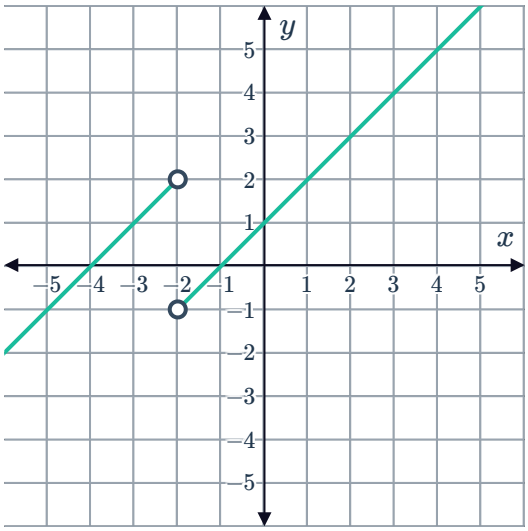
x	2.9	2.99	2.999	3	3.001	3.01	3.1
$\left(\dfrac{e^{x-3} + x - 4}{x - 3} \right)$				–			

b Does the above limit exist? Explain your answer.

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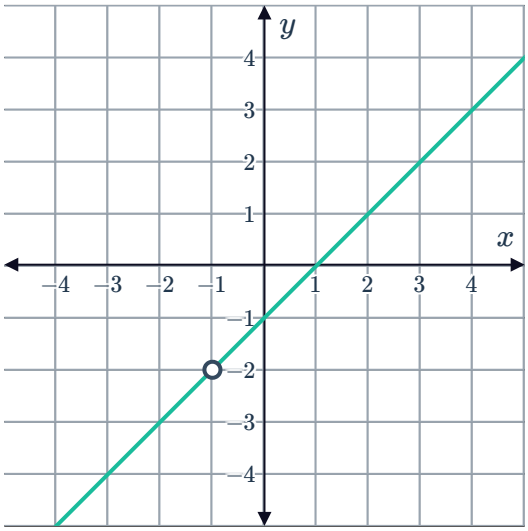
Graphs

9 Consider the graph of the piecewise function $g(x)$:



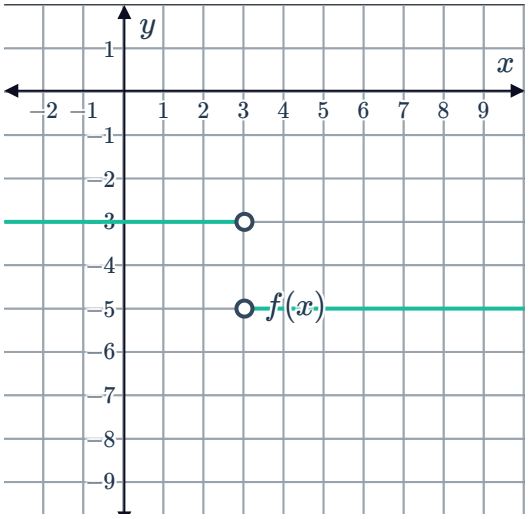
- **a** If we start at -4 and move along $g(x)$ to the right towards $x = -2$, what y-value do we approach?
- **b** If we start at 0 and move along $g(x)$ to the left towards $x = -2$, what y-value do we approach?
- **c** Does the $\lim_{x \to -2} g(x)$ exist? Explain your answer.

10 Consider the graph of the function $f(x) = \dfrac{x^2 - 1}{x + 1}$:



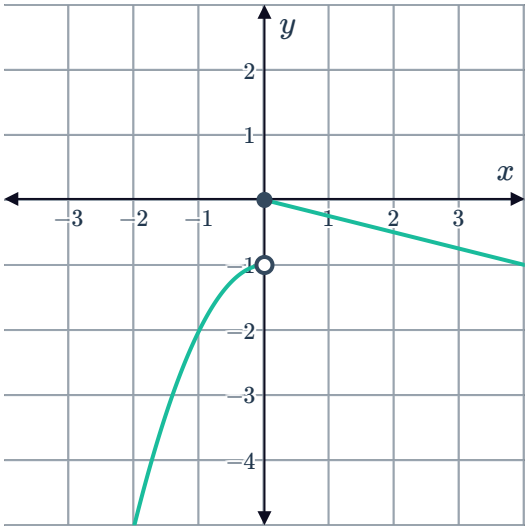
- **a** If we start at $x = -3$ and move along the function to the right towards $x = -1$, what y-value do we approach?
- **b** If we start at $x = 1$ and move along the function to the left towards $x = -1$, what y-value do we approach?
- **c** Describe the behaviour of $f(x) = \dfrac{x^2 - 1}{x + 1}$ as x approaches -1.
- **d** Write part (c) using limit notation.

11 Consider the graph of $f(x)$:
Does the limit $\lim_{x \to 3} f(x)$ exist? Explain your answer.

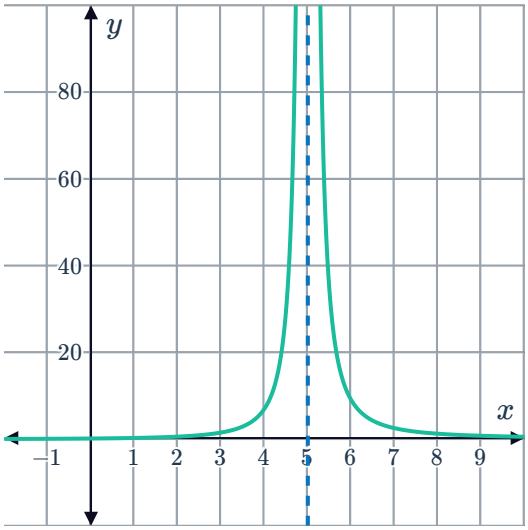


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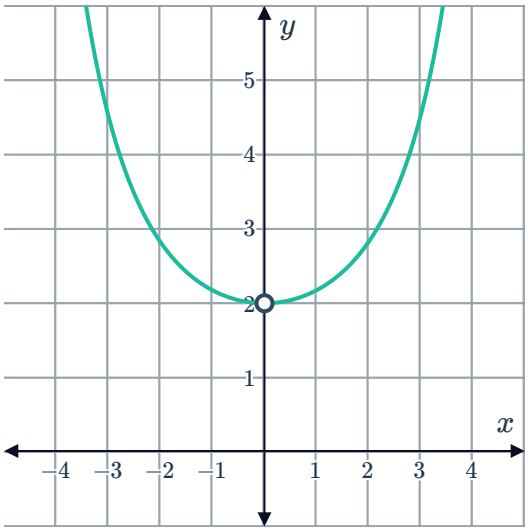
12 Consider the graph of $f(x)$:
Does the limit $\lim_{x \to 0} f(x)$ exist? Explain your answer.



13 Consider the graph of $y = \dfrac{x+3}{(x-5)^2}$:
Does the limit $\lim_{x \to 5} \left(\dfrac{x+3}{(x-5)^2} \right)$ exist?
Explain your answer.

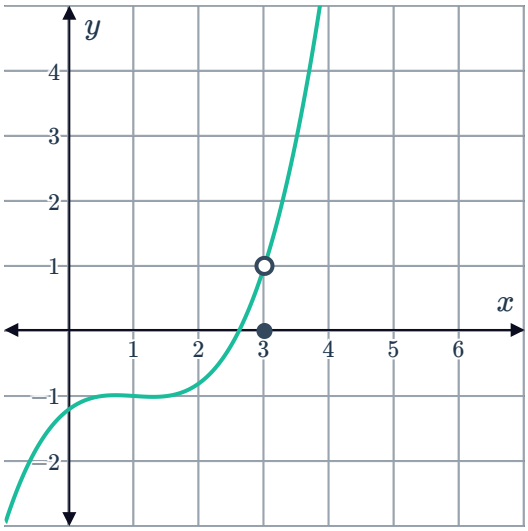


14 Consider the graph of $y = \dfrac{x^2}{1-\cos x}$:
Does the limit $\lim_{x \to 0} \left(\dfrac{x^2}{1-\cos x} \right)$ exist?
Explain your answer.

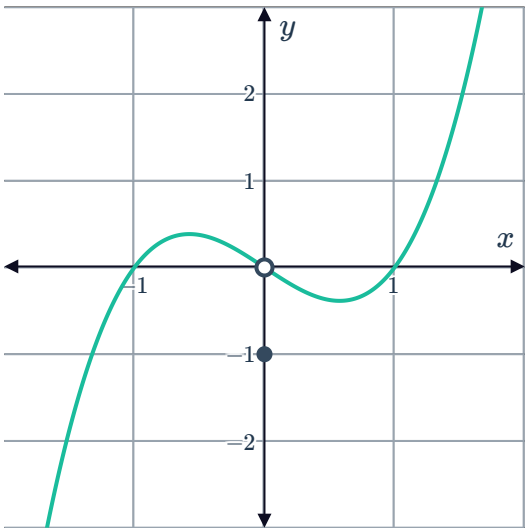


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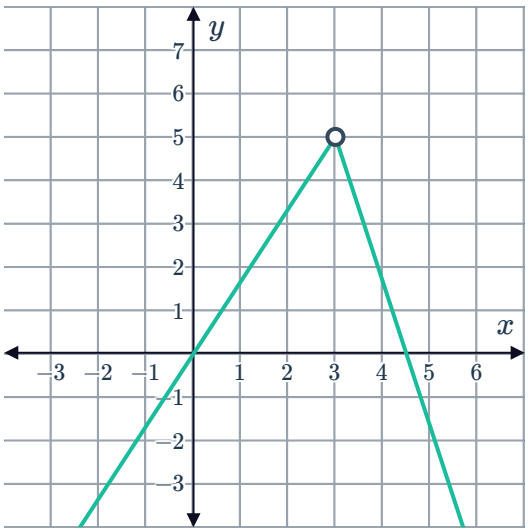
15 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 3} f(x)$.



16 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 0} f(x)$.

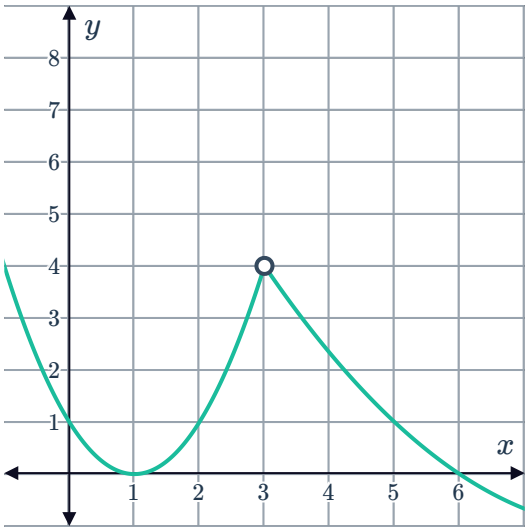


17 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 3} f(x)$.

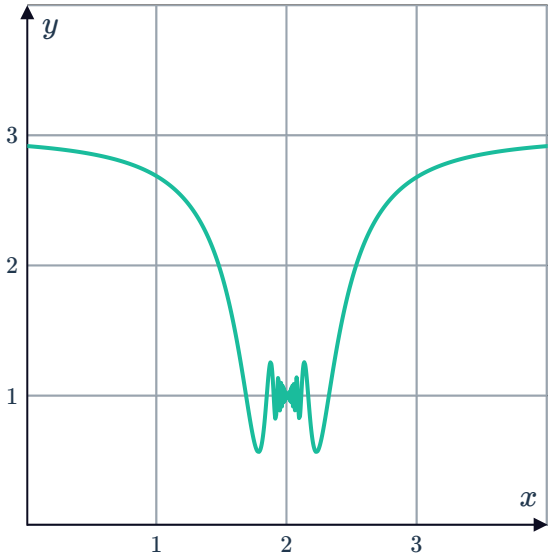


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18 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 3} f(x)$.



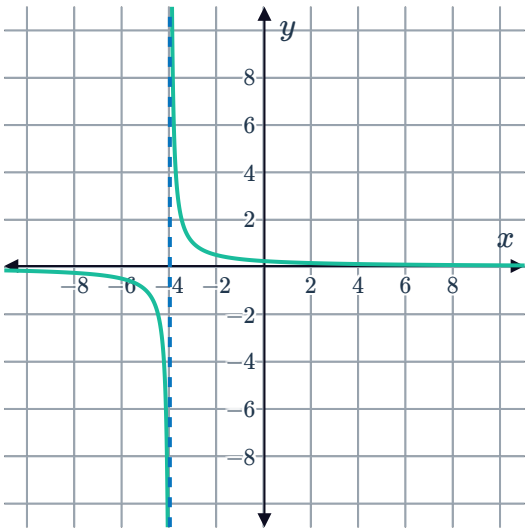
19 Consider the graph of the function $f(x)$:
Find the value of $\lim_{x \to 2} f(x)$.



20 Consider the graph of the function $f(x) = \dfrac{1}{x+4}$:
Find the value of:

a $\lim_{x \to -4^+} \left(\dfrac{1}{x+4} \right)$

b $\lim_{x \to -4^-} \left(\dfrac{1}{x+4} \right)$



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